

Mayank Singh

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Summary

- Motivated B.Tech student at Indian Institute of Technology IIT (BHU) Varanasi seeking a software development internship to leverage strong programming skills in C++, along with experience in Full-stack development using React, Node.js, Express, and MongoDB, and knowledge in computational finance and Machine Learning .
- Eager to contribute to innovative projects, enhance technical expertise, and collaborate in a dynamic team environment.

Skills

- **Languages:** C++, JavaScript (ES6+), Python
- **Frameworks & Tools:** React, Node.js, Express, MongoDB (Mongoose), REST APIs, WebSockets (Socket.IO), SQL, PostgreSQL, Git & GitHub, NumPy, Pandas
- **Other Skills:** Software Development, Full-stack Development, Database Management, Data Structures & Algorithms, Object-Oriented Programming (OOP), Unit Testing, Debugging & Performance Optimization

Education

2023–27	B.Tech , Indian Institute of Technology (BHU), Varanasi	<i>CGPA: 8.22</i>
2022	Class 12 , CBSE Shiv Jyoti International School	<i>90.6%</i>
2020	Class 10 , CBSE St. Xavier's School	<i>95.80%</i>

Achievements

- **Competitive Programming** — Solved 800+ problems across **Codeforces**, **LeetCode**, and **GeeksforGeeks** and achieved a peak Codeforces rating of **1207**. Regularly participated in rated contests with strengths in dynamic programming, graph, greedy algorithms, and binary search.
- **JEE Achievements** — Cleared **JEE Main** and **JEE Advanced** with a strong 99 percentile to secure admission into IIT (BHU), demonstrating analytical and problem-solving aptitude.
- **Campus & Startup Involvement** — Presented a mentorship platform at **Tech Startup Weekend (E-Cell)**, collaborating with peers to design an MVP for student–mentor pairing. Gained experience in product thinking, pitching, rapid prototyping, full-stack development, and delivering a functional prototype under tight deadlines.

Projects

Monte Carlo Simulation — Asian Option Pricer [GitHub Repo](#)

- Built a Monte Carlo pricer for Asian call options using historical data fetched via yfinance, supporting single and multi-ticker comparison modes.
- Engineered vectorized path simulations, convergence checkpointing, and CSV exports to ensure statistical robustness and runtime efficiency, achieving results within a **95% confidence interval**.
- **Tech Stack:** Python, NumPy, Pandas, Matplotlib, yfinance, Streamlit

Web-Chat App — Nodetalk [GitHub Repo](#)

- Developed a full-stack real-time chat application using React, Node.js, Express, and MongoDB, implementing Socket.IO and REST APIs with JWT authentication .
- Supported 150 concurrent WebSocket connections and 1,200 messages per day; deployed the full-stack app on Render (<https://nodetalkapp-3.onrender.com>).
- **Tech Stack:** React, Node.js, Express, Socket.IO, MongoDB (Mongoose), HTML, CSS, JavaScript

Extracurricular Activities

- **Chess** — Active on [Chess.com](#) ([singhmayank6386](#)) with peak Rapid rating of **1393**; developed strategic thinking, analytical reasoning, and problem-solving skills.